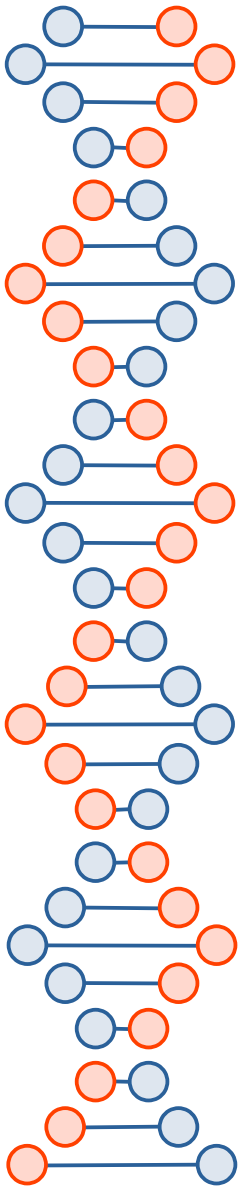


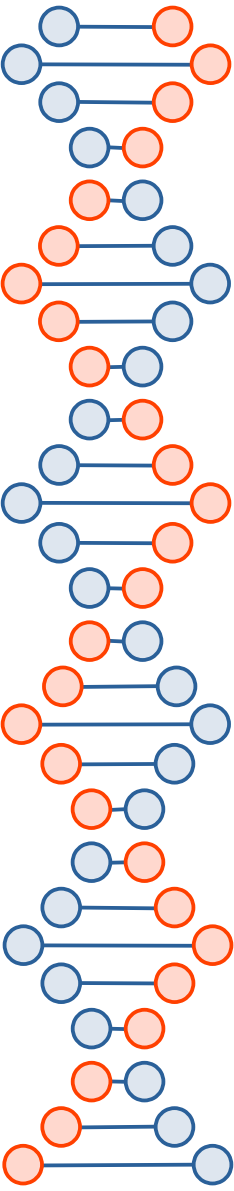
Métricas de evaluación de modelos de regresión: Error Absoluto Medio y Raíz del Error Cuadrático Medio

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Machine Learning



Outline

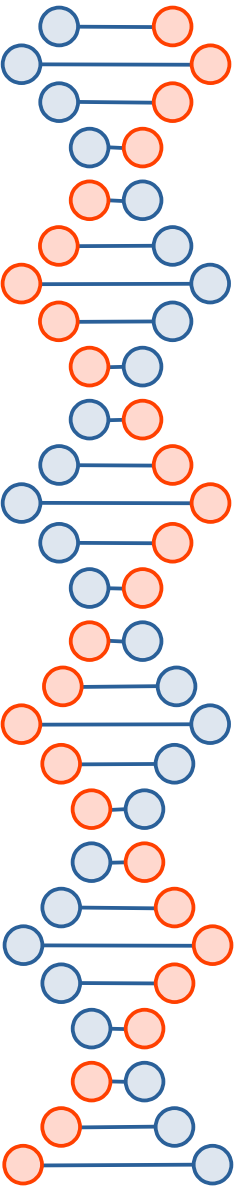
- How to implement Mean Absolute Error for regression.
- How to implement Root Mean Squared Error for regression.



Mean Absolute Error (1)

- Regression problems are those where a real value is predicted.
- An easy metric to consider is the error in the predicted values as compared to the expected values.
- The Mean Absolute Error or MAE for short is a good first error metric to use.
- It is calculated as the average of the absolute error values, where absolute means made positive so that they can be added together.

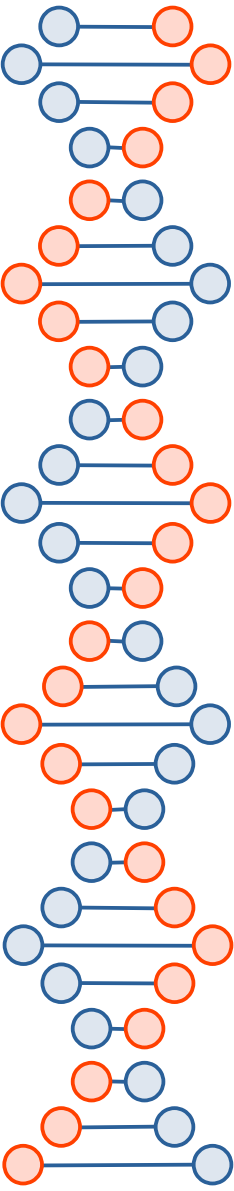
$$MAE = \frac{\sum_{i=1}^n abs(predicted_i - actual_i)}{\text{total predictions}}$$



Mean Absolute Error (2)

- Below is a function named `mae_metric()` that implements this metric.
- As above, it expects a list of actual outcome values and a list of predictions.
- We use the built-in `abs()` function to calculate the absolute error values that are summed together.
- Only one prediction (0.5) is correct, whereas all other predictions are wrong by 0.01.
- Therefore, we would expect the mean absolute error (or the average positive error) for these predictions to be a little less than 0.01.
- Below is an example that tests the `mae_metric()` function with the contrived dataset.
- Running this example prints the output below. We can see that as expected, the MAE was 0.008, a small value slightly lower than 0.01.

Mean Absolute Error (3)



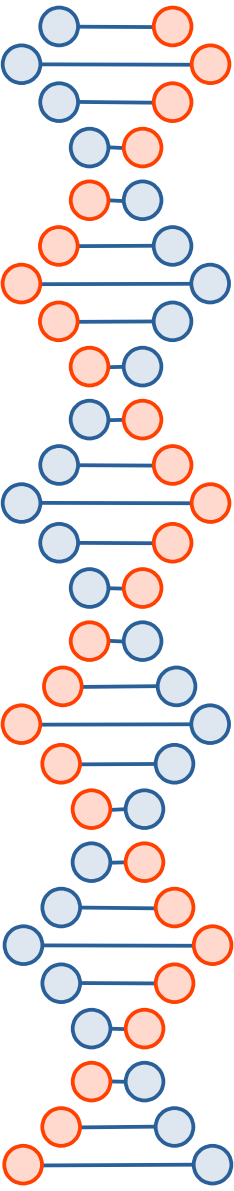
```
# Example of Calculating Mean Absolute Error
# Defined in Section 4.2.3 Mean Absolute Error¶
# Function To Calculate Mean Absolute Error.
# Calculate mean absolute error
sub mae_metric{
  my ($actual, $predicted) = @_ ;
  my $sum_error = 0.0;
  for my $i (0 .. $#actual){
    $sum_error += abs($predicted->[$i] - $actual->[$i]);
  }
  return sprintf '%0.3f', $sum_error / @actual;
}

# Small Set of Contrived Regression Predictions and Actual Values.
$actual = [0.1, 0.2, 0.3, 0.4, 0.5];
$predicted = [0.11, 0.19, 0.29, 0.41, 0.5];
print "actual\tpredicted\n";
printf "%s\t%s\n", $_->[0], $_->[1] for zip ($actual, $predicted);
# Test MAE
my $mae = mae_metric($actual, $predicted);
print dump $mae;
# Example Output From Calculating the Mean Absolute Error.
# 0.008
```

Root Mean Squared Error (1)

- Another popular way to calculate the error in a set of regression predictions is to use the Root Mean Squared Error.
- Shortened as RMSE, the metric is sometimes called Mean Squared Error or MSE, dropping the Root part from the calculation and the name.
- RMSE is calculated as the square root of the mean of the squared differences between actual outcomes and predictions.
- Squaring each error forces the values to be positive, and the square root of the mean squared error returns the error metric back to the original units for comparison.

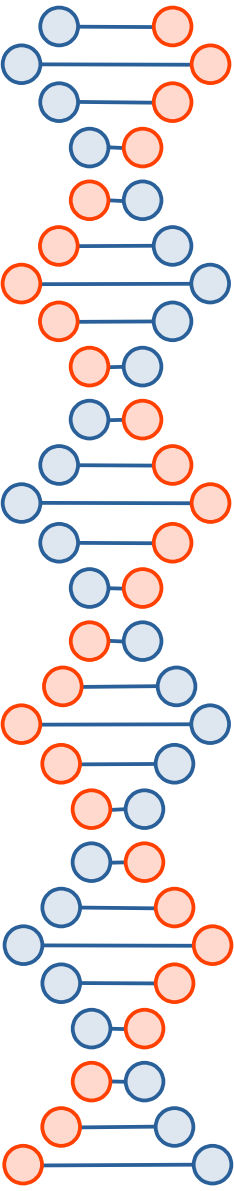
$$RMSE = \sqrt{\frac{\sum_{i=1}^n (\text{predicted}_i - \text{actual}_i)^2}{\text{total predictions}}}$$



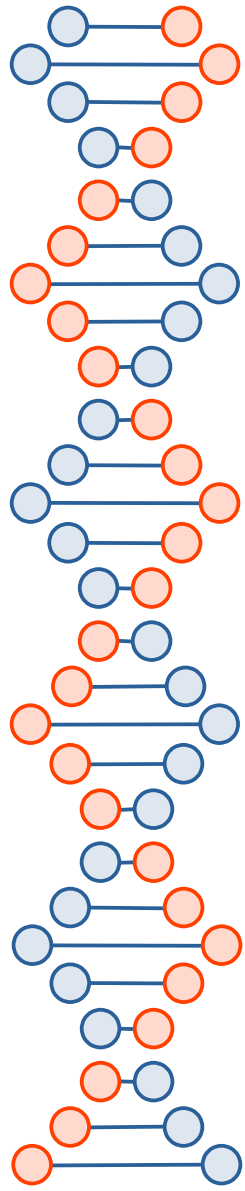
Root Mean Squared Error (2)

- Below is an implementation of this in a function named `rmse_metric()`.
- It uses the `sqrt()` function from the `math` module and uses the `**` operator to raise the error to the 2nd power.
- We can test this metric on the same dataset used to test the calculation of Mean Absolute Error above.
- Below is a complete example.
- Again, we would expect an error value to be generally close to 0.01.
- Running the example, we see the results below. The result is slightly higher at 0.0089.
- RMSE values are always slightly higher than MSE values, which becomes more pronounced as the prediction errors increase.
- This is a benefit of using RMSE over MSE in that it penalizes larger errors with worse scores.

Root Mean Squared Error (3)



```
# Example of Calculating the Root Mean Squared Error
# Defined in Section 4.2.4 Root Mean Squared Error
# Function To Calculate Root Mean Squared Error.
# Calculate root mean squared error
sub rmse_metric{
  my ($actual, $predicted) = @_ ;
  my $sum_error = 0.0;
  for my $i (0 .. $#actual){
    my $prediction_error = $predicted->[$i] - $actual->[$i];
    $sum_error += ($prediction_error ** 2);
  }
  my $mean_error = $sum_error / @actual;
  return sprintf '%0.4f', sqrt($mean_error);
}
# Test RMSE
$actual = [0.1, 0.2, 0.3, 0.4, 0.5];
$predicted = [0.11, 0.19, 0.29, 0.41, 0.5];
my $rmse = rmse_metric($actual, $predicted);
print dump $rmse;
# Example Output From Calculating the Root Mean Squared Error.
# 0.00894427191
```

Summary

- In this tutorial, you discovered how to implement algorithm prediction performance metrics from scratch. Specifically, you learned:
- How to implement and interpret mean absolute error for regression.
- How to implement and interpret root mean squared error for regression.